

RWorksheet_prado#3a

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# Q1a: Produce a vector that contains the first 11 letters
LETTERS[1:11]

## [1] "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K"

# Q1b: Produce a vector that contains the odd numbered letters (1, 3, 5, ..., 25)
LETTERS[seq(1, 26, by = 2)]

## [1] "A" "C" "E" "G" "I" "K" "M" "O" "Q" "S" "U" "W" "Y"

# Q1c: Produce a vector that contains the vowels
LETTERS[c(1, 5, 9, 15, 21)] # A, E, I, O, U

## [1] "A" "E" "I" "O" "U"

# Q1d: Produce a vector that contains the last 5 lowercase letters
letters[22:26] # V to Z

## [1] "v" "w" "x" "y" "z"

# Q1e: Produce a vector that contains letters between 15 to 24 letters in lowercase
letters[15:24]

## [1] "o" "p" "q" "r" "s" "t" "u" "v" "w" "x"

# Q2a: Create character vector for cities
city <- c("Tuguegarao City", "Manila", "Iloilo City", "Tacloban", "Samal Island", "Davao City")
city

## [1] "Tuguegarao City" "Manila"          "Iloilo City"      "Tacloban"
## [5] "Samal Island"    "Davao City"

# Q2b: Create numeric vector for temperatures
temp <- c(42, 39, 34, 34, 30, 27)
temp

## [1] 42 39 34 34 30 27

# Q2c: Create a dataframe to combine city and temp
df <- data.frame(city, temp)
df

##           city temp
## 1 Tuguegarao City  42
## 2      Manila    39
## 3   Iloilo City   34
## 4     Tacloban   34
## 5  Samal Island   30
## 6   Davao City   27
```

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# Q2e: Print the structure by using str() function
str(df)

## 'data.frame':    6 obs. of  2 variables:
## $ city: chr  "Tuguegarao City" "Manila" "Iloilo City" "Tacloban" ...
## $ temp: num  42 39 34 34 30 27

# Q2f: Content of row 3 and row 4
df[3:4, ]

##           city temp
## 3 Iloilo City   34
## 4  Tacloban    34

# Q2g: Display the city with highest and lowest temperature

# City with highest temperature (using which.max)
highest_temp_city <- df[which.max(df$Temperature), ]

# City with lowest temperature (using which.min)
lowest_temp_city <- df[which.min(df$Temperature), ]

cat("City with Highest Temperature:\n")

## City with Highest Temperature:
print(highest_temp_city)

## [1] city temp
## <0 rows> (or 0-length row.names)
cat("\nCity with Lowest Temperature:\n")

##
## City with Lowest Temperature:
print(lowest_temp_city)

## [1] city temp
## <0 rows> (or 0-length row.names)

# Q2a: Create a matrix of one to eight (1:8) and eleven to fourteen (11:14)
# Data sequence: 1, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14 (total 12 elements)
# Dimensions: 4 columns and 3 rows
data_vector <- c(1:8, 11:14)
matrix_a <- matrix(data_vector, nrow = 3, ncol = 4)
matrix_a

##      [,1] [,2] [,3] [,4]
## [1,]    1    4    7   12
## [2,]    2    5    8   13
## [3,]    3    6   11   14

# Q2b: Multiply the matrix by two
matrix_b <- matrix_a * 2
matrix_b

##      [,1] [,2] [,3] [,4]
## [1,]    2    8   14   24

```

```
## [2,] 4 10 16 26
## [3,] 6 12 22 28
```

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# Q2c: What is the content of row 2?
# The output is the R code's result.
matrix_a[2, ]
```

```
## [1] 2 5 8 13
```

```
# Q2d: Display column 3 and column 4 in row 1 and row 2
matrix_a[1:2, 3:4]
```

```
##      [,1] [,2]
## [1,]    7   12
## [2,]    8   13
```

```
# Q2e: Display only the columns in 2 and 3, row 3
matrix_a[3, 2:3]
```

```
## [1] 6 11
```

```
# Q2f: Display only column 4
matrix_a[, 4]
```

```
## [1] 12 13 14
```

```
# Q2g: Name rows as isa, dalawa, tatlo and columns as uno, dos, tres, quatro
rownames(matrix_b) <- c("isa", "dalawa", "tatlo")
colnames(matrix_b) <- c("uno", "dos", "tres", "quatro")
matrix_b
```

```
##      uno dos tres quatro
## isa      2  8  14      24
## dalawa   4 10  16      26
## tatlo    6 12  22      28
```

```
# Q3a: Create an array with the values repeated twice, 4 columns, 2 rows, 3 dimensions
values_3a <- c(1, 2, 3, 6, 7, 8, 9, 0, 3, 4, 5, 1) # 12 unique values
data_3a <- rep(values_3a, 2) # Total 24 elements
# dim = c(rows, columns, dimensions) = c(2, 4, 3)
array_3a <- array(data_3a, dim = c(2, 4, 3))
array_3a
```

```
## , , 1
##
##      [,1] [,2] [,3] [,4]
## [1,]    1    3    7    9
## [2,]    2    6    8    0
##
## , , 2
##
##      [,1] [,2] [,3] [,4]
## [1,]    3    5    1    3
## [2,]    4    1    2    6
##
## , , 3
##
##      [,1] [,2] [,3] [,4]
## [1,]    7    9    3    5
```

```
## [2,]      8      0      4      1
#Q3bIt has 3 dimensions

# Q3c: Name rows, columns, and dimensions

# Row names (2 rows)
row_names <- letters[1:2] # a, b

# Column names (4 columns)
col_names <- LETTERS[1:4] # A, B, C, D

# Dimension names (3 dimensions)
dim_names <- c("1st-Dimensional Array", "2nd-Dimensional Array", "3rd-Dimensional Array")

# Apply names
dimnames(array_3a) <- list(row_names, col_names, dim_names)
array_3a

## , , 1st-Dimensional Array
##
##   A B C D
## a 1 3 7 9
## b 2 6 8 0
##
## , , 2nd-Dimensional Array
##
##   A B C D
## a 3 5 1 3
## b 4 1 2 6
##
## , , 3rd-Dimensional Array
##
##   A B C D
## a 7 9 3 5
## b 8 0 4 1
```